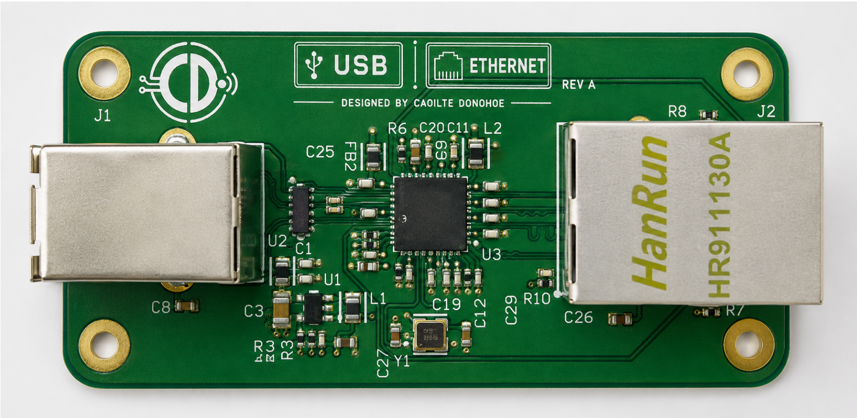
PCB, IoT Solutions & Embedded Engineering Portfolio

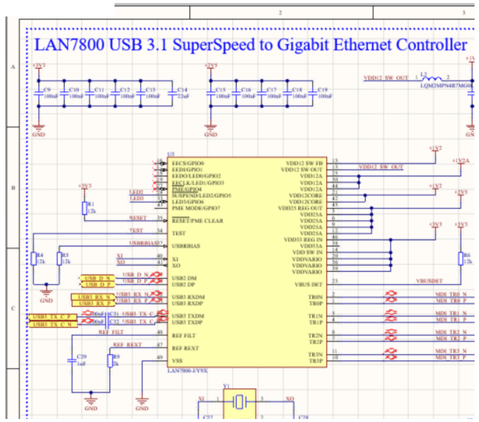
USB 5 Gbit/s to Gigabit Ethernet Adapter - LAN7800 | 4-layer high-speed PCB

4-layer LAN7800 USB 5 Gbit/s to Gigabit Ethernet adapter designed from schematic capture through PCB layout and fabrication outputs.

PCB Assembly



Schematic excerpt



BOM excerpt

Ref	Qty	Description	Manufacturer	MPN
U3	1	USB 3.x to Gigabit Ethernet controller	Microchip	LAN7800
J2	1	RJ45 with integrated magnetics	HanRun	HR911130A

Design deliverables

- Schematic capture and component integration
- 4-layer PCB layout with controlled USB and Ethernet differential routing
- Rule-driven DRC and fabrication documentation
- Gerber X2, NC Drill and editable design source

WHAT I DELIVER

IoT Solutions Design

Schematic -> PCB -> Manufacturing

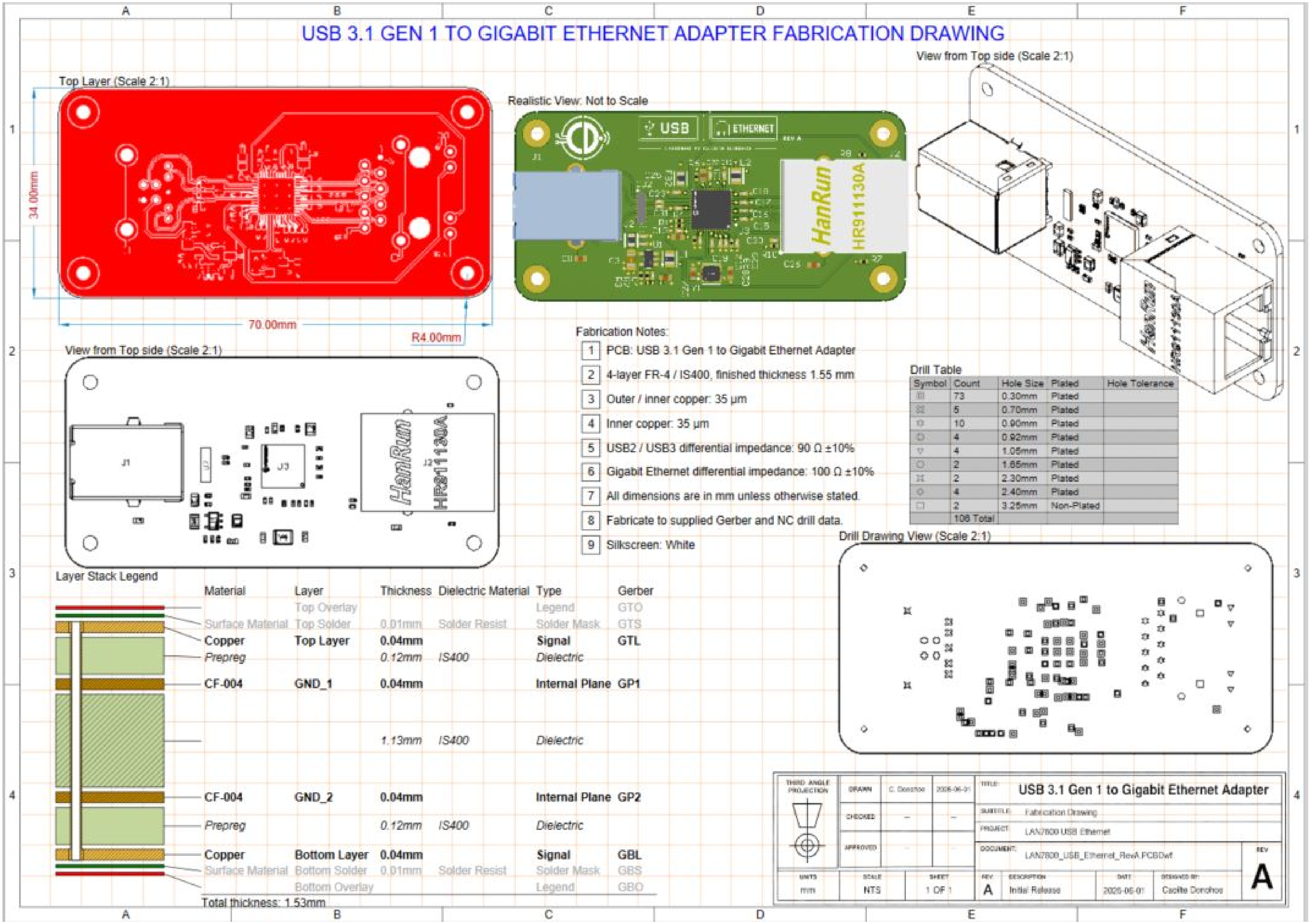
High-speed / RF / mixed-signal design

Production-ready outputs

SECTORS Industrial | Medical Devices | Environmental Monitoring | Energy | Consumer / Connected Products

From PCB Layout to Fabrication

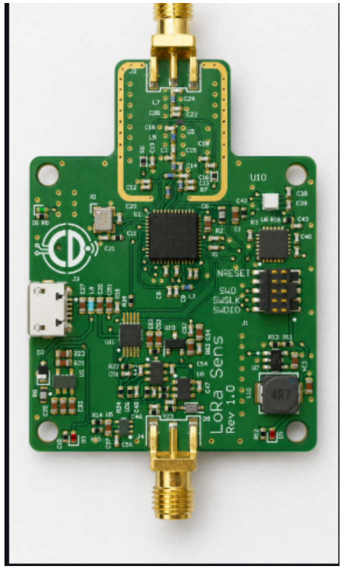
Dimensioned PCB, layer stack, drill information and fabrication documentation



Design Verification, RF & Release Package

Manufacturing outputs together with IoT, RF/wireless and embedded design capability

RF, Wireless & IoT Hardware Design



STM32WL LoRa Sensor Board

Mixed-signal wireless IoT sensor design using STM32WL with integrated sub-GHz radio, RF matching/filtering, antenna interface and deliberate separation of RF, digital and analog sections.

- IoT sensor-node / LoRa architecture
- RF matching and filtering
- Antenna / SMA interface
- Mixed-signal PCB partitioning

IoT **Sub-GHz** **LoRa**

PCB design release package

Deliverable	Output
Schematic	Editable design + schematic PDF
PCB layout	Routed multi-layer PCB design
BOM	Manufacturer and part-number list
Placement	Component placement / assembly data
Fabrication	Gerber X2 + NC Drill
Drawing	Fabrication / assembly documentation
Verification	Final design-rule checks

Additional engineering support

Beyond PCB and IoT hardware design, I can also support board bring-up, embedded firmware integration and circuit simulation.

Board bring-up & debug

Initial power-up, interface checks, fault finding and hardware verification

Firmware bring-up

MCU peripheral setup, board-level firmware integration and hardware/firmware debug

Circuit simulation

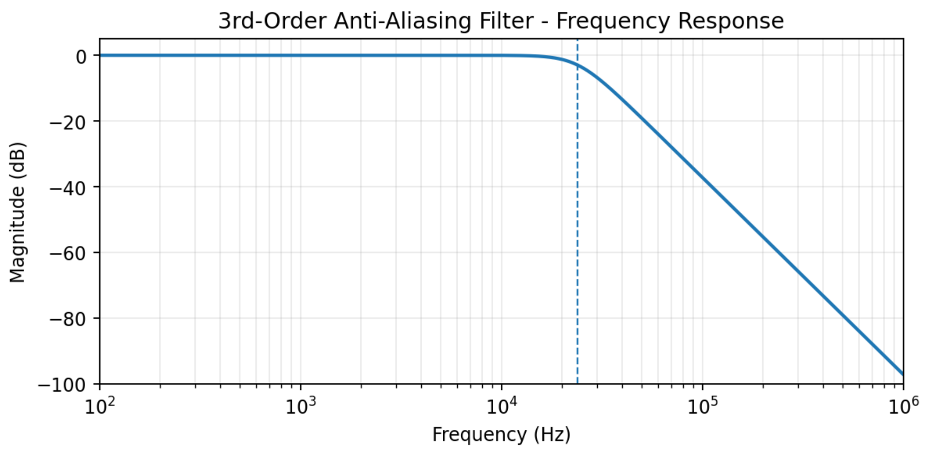
Power, analog and signal-path simulation to support design decisions

Circuit Simulation & Engineering Analysis

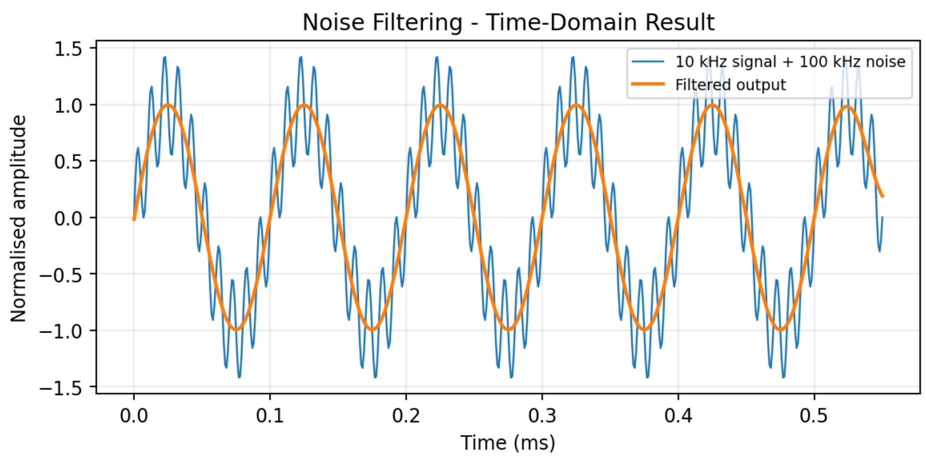
Examples from the STM32WL LoRa IoT sensor-board design - LTspice and Python/Jupyter analysis

Simulation is used where it adds value: checking analog filtering, signal behaviour and RF trade-offs before committing the design to PCB.

Analog filter response



Noise filtering - time domain



Analog / ADC simulation

- Sallen-Key anti-aliasing filter analysis
- Bias / VCOM generation checks
- Transient and pulse-response analysis
- Pseudo-differential ADC signal-path investigation
- Python/Jupyter used as an independent frequency/time-domain check

STM32WL RF output-network simulation

- 868 MHz output matching and harmonic-filter trade-off analysis
- RF network simulation used to compare matching/filter options
- Insertion loss and harmonic rejection considered together
- Simulation supported component and filter-architecture decisions
- Used as design support before PCB implementation